

1) Transactions:

```
SELECT m.name AS medicine_name, m.rating AS medicine_rating,
COUNT(tb.symptom_name) AS symptoms_treated FROM
stage2_schema.medicines m JOIN stage2_schema.treatedBy tb ON m.name =
tb.medicine_name JOIN stage2_schema.currentlyHas ch ON tb.symptom_name
= ch.symptom_name WHERE ch.user_email = ? GROUP BY m.name, m.rating
ORDER BY symptoms_treated DESC, m.rating DESC LIMIT 15; SELECT p.id
AS pharmacy_id, p.address, p.city, p.state, COUNT(DISTINCT
c.medicine_name) AS num_medicines_carried FROM
stage2_schema.pharmacies p JOIN stage2_schema.carries c ON p.id =
c.pharmacy_id JOIN stage2_schema.treatedBy tb ON c.medicine_name =
tb.medicine_name JOIN stage2_schema.currentlyHas ch ON tb.symptom_name
= ch.symptom_name WHERE ch.user_email = ? GROUP BY p.id, p.address,
p.city, p.state ORDER BY num_medicines_carried DESC LIMIT 15;
```

2) Triggers:

```
CREATE TRIGGER `limit_currentlyHas` BEFORE
INSERT
ON
`stage2_schema`.`currentlyHas` FOR EACH ROW
BEGIN
DECLARE
cnt INT;
SELECT
COUNT(*) INTO cnt
FROM
`stage2_schema`.`currentlyHas`
WHERE
user_email = NEW.user_email;
IF
cnt >= 5 THEN SIGNAL SQLSTATE '45000'
SET
MESSAGE_TEXT = 'Error: A user cannot have more than 5 currentlyHas records.';
END IF
;
END
```

3) **PROCEDURE:**

```
CREATE PROCEDURE
stage2_schema.getMedicinesBySymptomsAdvanced(IN user_email_param
VARCHAR(255)) READS SQL DATA
BEGIN
CREATE TEMPORARY TABLE tempUserSymptoms AS
SELECT `symptom_name`
FROM `stage2_schema`.`currentlyHas`
WHERE `user_email` = user_email_param;

SELECT
t.`symptom_name`,
COUNT(DISTINCT ch.`user_email`) AS user_count
FROM
tempUserSymptoms t
JOIN
`stage2_schema`.`currentlyHas` ch
ON
t.`symptom_name` = ch.`symptom_name`
GROUP BY
t.`symptom_name`
ORDER BY
user_count DESC;
SELECT
m.`name` AS medicine_name,
m.`rating` AS medicine_rating,
COUNT(tb.`symptom_name`) AS symptoms_treated
FROM
`stage2_schema`.`medicines` m
JOIN
`stage2_schema`.`treatedBy` tb
ON
m.`name` = tb.`medicine_name`
JOIN
tempUserSymptoms t
ON
tb.`symptom_name` = t.`symptom_name`
GROUP BY
m.`name`,
m.`rating`
ORDER BY
symptoms_treated DESC,
medicine_rating DESC
LIMIT
15;
DROP
```

```
TEMPORARY TABLE tempUserSymptoms;  
END
```